

Election Analytics Platform for Portugal

Advanced Topics in Databases Practical Assignment

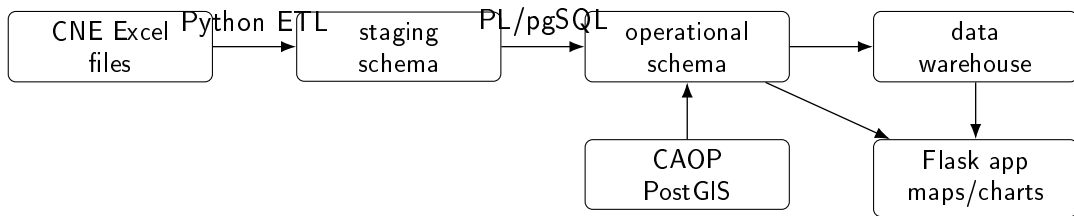
Rui Coelho, up202106772 Ricardo Ribeiro, up202104806

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Goal and Scope

- Database-centred platform for analysing Portuguese election results.
- Dataset: official CNE Autarquicas 2021 spreadsheet package.
- Spatial layer: official DGT CAOP GeoPackages loaded into PostGIS.
- Thin Flask frontend using explicit SQL through psycopg2.

Architecture



- Core entities: elections, organs, territories, candidacies.
- Result entities: turnout, vote results, seat results, elected members.
- Territory hierarchy supports district/region, municipality, and parish drill-down.
- Candidacies model parties, coalitions, and citizen groups through a typed local candidacy table.
- Keys, constraints, uniqueness rules, and indexes support integrity and dashboard queries.

- Rerunnable Python loader parses the CNE spreadsheets and populates staging.
- PL/pgSQL procedure transforms staging rows into normalized operational tables.
- Warehouse refresh rebuilds dimensions and facts from the operational schema.
- CAOP loader reads GeoPackages, reprojects to EPSG:3763, and updates territory geometries.

Dataset	Rows
Result rows	3719
Candidate votes	12277
Percentages and mandates	12277
Elected members parsed	35491

Warehouse and Analytical SQL

- Star schema: election, organ, territory, and candidacy dimensions.
- Central fact table stores votes, official percentages, mandates, and turnout measures.
- Analytical SQL includes:
 - D'Hondt mandate allocation;
 - three window-function queries;
 - ROLLUP and CUBE;
 - FILTER and JSON aggregation.

- SQL functions: vote share, turnout rate, D'Hondt allocation.
- PL/pgSQL procedures: load operational schema from staging; refresh warehouse.
- Triggers enforce meaningful business rules:
 - no negative turnout/vote/mandate values;
 - voters cannot exceed registered voters;
 - vote and seat rows must match the candidacy scope.

- PostGIS stores geometries as geometry(MultiPolygon, 3763).
- Loaded geometry coverage:
 - district/region: 20 / 20;
 - municipality: 308 / 308;
 - parish: 2949 / 3083.
- Frontend includes election/organ selection, interactive Leaflet municipality map, result table, turnout summary, vote-share chart, and seat chart.

- PostgreSQL validation checks staging, operational, warehouse, materialized views, and D'Hondt results.
- Agueda municipal council D'Hondt matches official mandates:
 - PPD/PSD.MPT: 4;
 - PS: 2;
 - CDS-PP: 1.
- Flask endpoint tests:
 - homepage returned HTTP 200;
 - municipality results returned HTTP 200;
 - map endpoint returned 308 municipality features.

Limitations and Extensions

- CAOP 2025 is used instead of CAOP 2021; it is official DGT data but not a perfect historical match.
- Parish geometries are partial because some codes changed between 2021 and 2025.
- Some elected-member rows do not match normalized parish candidacies exactly.
- Future work: historical comparisons, richer district dashboards, and improved parish reconciliation.

- ➊ Show the repository structure and SQL/ETL files.
- ➋ Run validation query for Agueda D'Hondt.
- ➌ Open the Flask map and select a municipality.
- ➍ Show result table, turnout, vote-share chart, seat chart, and elected members.